

Repository Reorganization & 3DPW Dataset Preparation

1. Directory Structure

Per the user's instructions, all documentation, research papers, proposals, notes, and codebase modules have been organized into the following clean hierarchy:

```

/home/dat/HMR/
├── documents/
│   ├── pdf/
│   │   └── # Formatted for human reading & review
│   │   ├── papers/
│   │   │   └── # Foundational & related research papers
│   │   │   ├── 2308.04079v1_compressed.pdf
│   │   │   ├── 2409.04196v2.pdf
│   │   │   ├── 2605.02784v2.pdf
│   │   │   │   └── # HumanSplatHMR
│   │   │   ├── 3603618.pdf
│   │   │   │   └── # HMR Survey & Benchmarks
│   │   │   └── frai-8-1709229.pdf
│   │   ├── notes/
│   │   │   └── # Implementation plans, reports, walkthroughs
│   │   │   ├── diffnorm_contact_hmr_implementation_plan.pdf
│   │   │   ├── walkthrough.pdf
│   │   │   ├── papers_analysis_report.pdf
│   │   │   └── repository_organization_and_dataset_setup.pdf
│   │   ├── proposals/
│   │   │   └── # Research proposals and comparative evaluations
│   │   │   ├── proposal.pdf
│   │   │   │   └── # DiffNorm-Contact HMR Full Proposal
│   │   │   ├── hmr_3dgs_critique_and_proposal.pdf
│   │   │   ├── comparison_proposal_vs_humansplathmr.pdf
│   │   │   ├── falsifiable_hmr_3dgs_proposal.pdf
│   │   │   └── deep_research_hmr_3dgs_proposals.pdf
│   └── md/
│       └── # Formatted for AI agent parsing & analysis
│       ├── papers/
│       │   └── 3603618.md
│       │       └── # Markdown transcription of survey & tables
│       ├── notes/
│       │   ├── diffnorm_contact_hmr_implementation_plan.md
│       │   ├── walkthrough.md
│       │   ├── AGENTS.md
│       │   │   └── # Project guidelines & agent instructions
│       │   └── repository_organization_and_dataset_setup.md
│       └── proposals/
│           ├── proposal.md
│           ├── hmr_3dgs_critique_and_proposal.md
│           └── comparison_proposal_vs_humansplathmr.md
├── code/
│   └── # Complete codebase
│   ├── src/
│   │   └── # Core geometry, physics, rendering, & pipeline
│   │   ├── geometry/
│   │   │   └── # SMPL wrapper, kinematic segments, mesh graph
│   │   ├── gaussian/
│   │   │   └── # Tangential disks & spatial covariances
│   │   ├── physics/
│   │   │   └── # Analytical Gaussian overlap  $K_{ij}$  & collision loss
│   │   ├── rendering/
│   │   │   └── # Screen-space normal rasterizer & loss functions
│   │   ├── optimization/
│   │   │   └── # Dual-frequency optimizer & gradient router
│   │   └── pipeline/
│   │       └── # 3DPW loader, feature cache, single-image optimizer
│   ├── scripts/
│   │   └── # Training, evaluation, preprocessing, & download
│   │   ├── download_3dpw_aria2c.sh
│   │   │   └── # High-speed parallel 3DPW aria2c downloader
│   │   ├── prepare_3dpw.py
│   │   │   └── # Extraction, verification, and sequence inventory
│   │   ├── train_colab.py
│   │   │   └── # Multi-epoch batched trainer for Google Colab T4
│   │   ├── preprocess_3dpw.py
│   │   │   └── # HDF5 foundation feature streamer
│   │   ├── eval_3dpw.py
│   │   │   └── # 3DPW benchmark evaluator (MPJPE, PA-MPJPE, PVE)
│   │   ├── eval_rich_contact.py
│   │   │   └── # RICH self-penetration evaluator ( $V_{pen}$ )
│   │   └── eval_cape_clothing.py
│   │       └── # CAPE loose garment evaluation runner
│   ├── tests/
│   │   └── # 14 automated unit tests (100% passing)
│   └── notebooks/
│       └── # Google Colab T4 interactive notebook
├── data/
│   └── # Dataset storage & caching
│   ├── downloads/
│   │   └── # Download archives (.zip) managed by aria2c
│   ├── 3dpw/
│   │   └── # Extracted sequenceFiles & imageFiles
│   └── cache/
│       └── # Precomputed DSINE normal maps & masks (HDF5)
├── models/
│   └── # SMPL body templates (models/smpl/SMPL_NEUTRAL.pkl)
└── checkpoints/
    └── # Trained model weights & optimization checkpoints

```

2. 3DPW Dataset aria2c Status & Extraction

The 3DPW dataset is downloaded directly from the official Max Planck Institute server

(<https://virtualhumans.mpi-inf.mpg.de/3DPW/>) using `aria2c` multi-connection segmented downloading (`-x 16 -s 16 -k 1M -c --file-allocation=none`):

1. `readme_and_demo.zip` (4.5 KB):

- Status: **Complete & Extracted** (`data/3dpw/readme.txt`).

2. `sequenceFiles.zip` (277 MB):

- Status: **100% Complete & Extracted** (`data/3dpw/sequenceFiles/`).
- Audit Results:
 - **Train Split:** 25 sequences, 17,914 frames (yielding 22,745 valid actor frame instances).
 - **Validation Split:** 12 sequences, 8,509 frames.
 - **Test Split:** 24 sequences, 26,240 frames.
 - **Total:** 61 sequences, 52,663 fully annotated video frames.
 - Formats verified: 72-dim SMPL axis-angle poses, 10-dim betas, 3-dim root translations, 4×4 camera extrinsics, and 3×3 intrinsic matrices.

3. `imageFiles.zip` (4.6 GB):

- Status: **100% Complete & Extracted** (`data/3dpw/imageFiles/`).
- Audit Results: 62 image sequence directories containing **53,806 RGB images** (`.jpg`).
- Download accelerated via `aria2c` segmented streaming, completing in ~35 minutes.

Automation Scripts & Loading Verification

- **Parallel Downloader Script:** [code/scripts/download_3dpw_aria2c.sh](#)
- **Automatic Unpacker & Inventory Auditor:** [code/scripts/prepare_3dpw.py](#)
- **Verified Dataset Loader:** [code/src/pipeline/dataset_3dpw.py](#) successfully loads live RGB frames, camera matrices, and aligned ground-truth SMPL parameters across all 22,745 training instances.

3. Verification

All 14 unit tests have been verified from both the repository root and `code/` :

```
PYTHONPATH=. pytest code/tests/ -v
```

All tests pass cleanly.